

High-Stiffness Control of Series Elastic Actuators Using a Noise Reduction Disturbance Observer

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Abstract—Rendering a high impedance on a series elastic actuator (SEA) can improve motion control performance while retaining collision safety by a lower physical stiffness. The safe impedance range can be increased with high-gain velocity feedback control, but, in practice, this is limited by noise. This article applies the noise reduction disturbance observer to inner-loop velocity control of an SEA, attenuating noise and allowing higher safe rendered stiffness compared with standard torque and torque/velocity hierarchical control. Closed-form expressions for maximum passive stiffness and Z -region are found, shown to depend strongly on the high-frequency noise gain, and used to optimize the control gains. Performance is experimentally verified on a reaction-force SEA, validating the passivity of the high-stiffness control in impact (free space and stiff environment) while rendering a safe stiffness 3.0 times the intrinsic stiffness.

Index Terms—High-stiffness control, impedance control, series elastic actuator (SEA), two mass system, Z -region.

I. INTRODUCTION

INTRINSIC compliance improves the safety of robots, which interact with uncertain environments or humans, but such robots have a range of application requirements for different phases of a task, including motion control, safe collision, and contact stability. This range of performance criteria requires a wide range of rendered impedance [1].

The impedance range a system can render is affected by many factors, including the physical stiffness, which is intentionally reduced in series elastic actuators (SEAs) [2]. The physical stiffness is an upper bound on rendered stiffness while maintaining spring port passivity [3], echoing results on motion control of flexible-joint robots [4]. However, recent work has shown that

coupled stable behavior beyond this limit can be achieved by considering passivity at another port [5].

The impedance range can be characterized by several metrics. In haptics, the range of impedance gains that are achievable is measured with Z -width [6], also extended to virtual inertias (M -width) [7]. In SEAs, an inner-loop controller also affects the rendered impedance (e.g., with low-gain inner-loop control, a high impedance may be commanded, but not achieved). This is addressed by the Z -region [8], which characterizes the area of realizable dynamics in the frequency domain: the depth of dynamics that an actuator can render over a width of frequencies. As low impedance is rarely problematic for SEAs, there is growing interest in high impedance on SEAs [9]–[11] and improving their motion control performance [12].

For many interactive systems, damping and derivative control are important for passivity and performance, which has been shown in haptics [13], flexible-joint robots [14], SEAs [10], [15], [16], and collision performance [17]. For SEAs, the use of velocity control as an innermost loop has also been shown to improve performance [18], but can increase the noise from position encoder differentiation [15]. Recent work has shown the ability to render stiffness higher than the physical stiffness [9], but is limited by differentiation noise [11]. While noise can be reduced by filtering, low-pass filters (LPFs) introduce phase lag, which is especially problematic in passivity-based stability arguments, as passivity can be viewed as a limit on absolute phase.

To reduce noise in high-performance motion control, the noise reduction disturbance observer (NRDOB) has been proposed [19]. The NRDOB builds on the architecture of the disturbance observer (DOB) [20], explicitly addressing the attenuation of sensor noise at high frequencies. The NRDOB reduces noise by transitioning from real system output (i.e., sensor measurements) to the output of the dynamic model at high frequencies. The NRDOB is here applied to reduce noise in the input to the velocity control loop, allowing higher velocity control gains to be realized.

The DOB (without noise reduction) is well established in SEA control [21]–[24]. It has been shown to improve torque-loop bandwidth and tracking [21] and reduce the need for other high-gain feedback control [24]. In the experiments in [11], the conventional DOB improved the performance of high-stiffness control of the SEA but only by a narrow margin.

Safety, in the sense of coupled stability with an unmodeled environment, also sets limitations in the control and performance. Passivity is the classical approach in impedance control [25], and passivity at the spring port of the SEA has been used to derive

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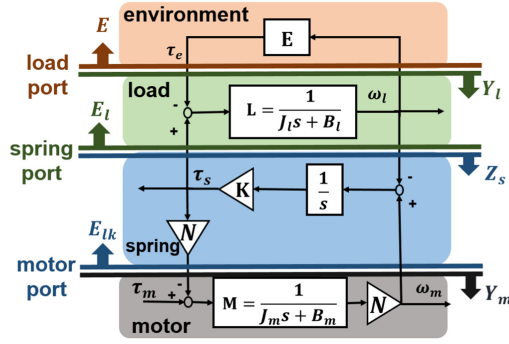


Fig. 1. SEA as a two-mass system, where ω_m and ω_l are motor and load velocities, respectively; M and L are motor and load dynamics, respectively; J_m , J_l , B_m , and B_l are inertia and damping values of the motor and the load, respectively; N is the reciprocal of the gear ratio; τ_m , τ_s , and τ_e are the motor torque, spring force, and external force, respectively; and K is the spring stiffness.

bounds on controller gains [3]. Subsequent work has considered passivity in various SEA models and control architectures [26], [27]. The conservatism of passivity is well established, leading authors to consider coupled stability over a restricted set of environment dynamic models [10], [21], [28], passivity at another port [5], [11], or considering a mixed-passivity/small-gain condition [29].

In this article, the goal of increasing the Z -region and maximum stiffness of an SEA is pursued. A stiffness impedance control is used with hierarchical proportional–derivative (PD) torque/velocity control (P-PD-PD), and the NRDOB is added to this, as presented in Section II. Closed-form expressions for the maximum stiffness and Z -region, based on load port passivity, are developed in Section III. In Section IV, the noise characteristics are analyzed, and controller gains optimized while keeping a fixed high-frequency noise gain. The controllers are compared experimentally in Section V, and the NRDOB achieves a safe stiffness 3.0 times higher than the intrinsic stiffness.

II. SEA MODEL AND CONTROL

This section introduces the dynamic model for an SEA and the control architectures compared here.

A. Dynamic Model

An SEA includes a motor, gearbox, a spring, and load [22], [23]. As shown in Fig. 1, the SEA can be dynamically modeled using a two-inertia model; the motor and the load are connected by the spring.

An SEA is a multi-input multi-output (MIMO) system with outputs including the positions or velocities of motor and load sides and the spring deformation. Here, two encoders measure all the outputs, and the following relationship is assumed for SEA outputs: $\tau_s = Ks^{-1}(\omega_m - \omega_l)$. From Fig. 1, the transfer functions of the MIMO SEA system are computed as

$$\omega_m = \frac{MN(1 + Ks^{-1}L)\tau_m - Ks^{-1}LMN\tau_e}{D(s)} \quad (1)$$

$$\tau_s = \frac{Ks^{-1}MN\tau_m + Ks^{-1}L\tau_e}{D(s)} \quad (2)$$

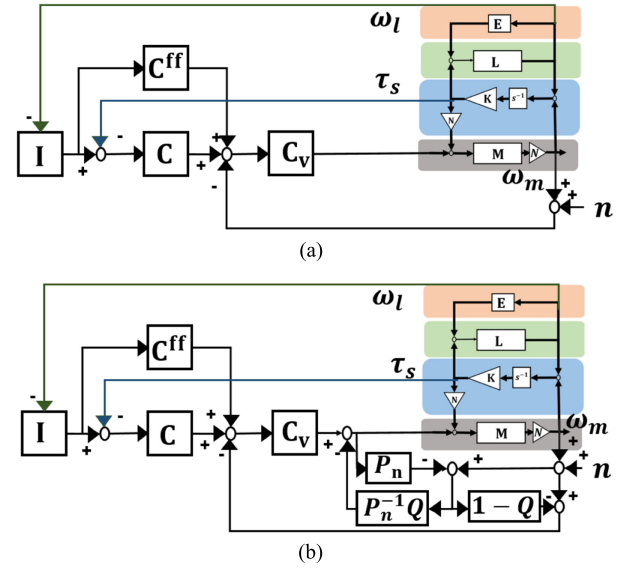


Fig. 2. Control structures for SEA high-stiffness control, where $I = K_{\text{imp}}/s$, P_n is nominal motor dynamics, C is the torque controller, C_v is the velocity controller, and C^{ff} is the feedforward control, and Q is a tuning filter for the NRDOB (LPF). (a) P-PD-PD control. (b) NRDOB with P-PD-PD.

$$\omega_l = \frac{Ks^{-1}L^2M\tau_m - L(1 + Ks^{-1}MN^2)\tau_e}{D(s)} \quad (3)$$

where $D(s) = 1 + Ks^{-1}L + Ks^{-1}MN^2$.

B. Control Architecture

This article considers the hierarchical control of SEAs, as seen in Fig. 2, where an outer loop impedance controller I generates a desired torque τ_d . The P-PD-PD controller has an inner-loop torque control and inner-most velocity control loop, as seen in Fig. 2(a) and used in [3] and [15].

The impedance controller is considered as a pure stiffness (proportional–P). The inner-loop control has been shown to be more important than outer-loop control [11]. The two inner-loop controllers are chosen as PD control, as common in SEA control [3]. Other proportional–integral–derivative variations are considered initially, but the chosen control configuration outperforms the others in terms of high-stiffness control.

1) Noise Reduction Disturbance Observer: A DOB can be closed around the motor dynamics (DOBm) [21], [30] or torque (DOBT) [23] and can improve the torque tracking performance and bandwidth. But in [11], no DOB, DOBm, and DOBT achieved $1.7K$, $1.8K$, and $1.65K$ in the experiments, respectively, which shows the limited high-impedance capabilities of the DOB. Theoretically, DOBs do not have any significant effect or merit in terms of passivity. The advantages of the DOB in high-stiffness control can be realized in practical applications, such as: 1) nominalization of an uncertain system and 2) compensation of undesired dynamics affecting the system, e.g., friction. These features of the DOB are well studied on SEAs [21], [24], [30] and are not discussed further in this article.

The NRDOB controller employs a noise-reducing DOB [19], which shares the DOB architecture—using a nominal model P_n and tuning filter Q (typically first- or second-order LPF) to

TABLE I
SPRING PORT DYNAMICS WITH FEEDFORWARD CONTROL OF THE TWO ARCHITECTURES ON A TWO-INERTIA SEA MODEL

	Spring port impedance	Spring port impedance after feed-forward control
P-PD-PD	$\frac{Ks^{-1}(MNC_v(C+C_2^{ff}))K_{imp}s^{-1}+(1+MNC_v)}{(1+MNC_v)+Ks^{-1}MN(C_vC+N)}$	$\frac{K_{imp}}{s} + \frac{(K-K_{imp})(1+MNC_v)}{s(1+MNC_v)+KMN(C_vC+N)}$
w/ NRDOB	$\frac{Ks^{-1}(MNC_v(C+C_3^{ff}))K_{imp}s^{-1}+(1+MNC_v)}{(1+MNC_v)+Ks^{-1}NM(C_vC+N(1+MNC_v)(1-Q))}$	$\frac{K_{imp}}{s} + \frac{(K-K_{imp})(1+MNC_v)}{s(1+MNC_v)+KMN(C_vC+N(1+MNC_v)(1-Q))}$

estimate then directly compensate unmodeled input. However, the NRDOB also modifies the output of the system above the bandwidth of the Q -filter to reduce the noise. It replaces the true output of the system with the model output, providing a smoothed output, which allows a higher derivative gain. Critically, it can avoid the need for simple LPFs, which typically increase phase, as phase is critical to passivity. Model-based filters such as the Kalman filter may reduce the filter phase lag and have been used for force estimation in SEAs [31], but not for motor state estimation, which is a topic for future work. The proposed high-stiffness control is developed by cascading an inner motor velocity loop and employing an NRDOB, as shown in Fig. 2(b).

When $|Q| \approx 1$, at low frequencies, the output response of the NRDOB is the same as the conventional DOB as

$$\omega_m(s) = \frac{P_n C_v}{1 + P_n C_v} \tau_m^d(s) \quad (4)$$

where C_v is the motor controller and the reference signal $\tau_m^d(s)$ (the output of C and C^{ff}). Referring to [19, eq. (7)], the main difference of the NRDOB is at high-frequency response has been modified to able to be controlled by the Q -filter design

$$\omega_m(s) = \frac{PQ}{P_n} n(s) \quad (5)$$

where P is the actual plant dynamics and $n(s)$ is the noise affecting $\omega_m(s)$. Notice that Q of the NRDOB can influence the transfer function between the output ω_m and the noise n .

Throughout this article, the NRDOB is applied to the motor loop, as shown in Fig. 2(b). Hence, the plant is denoted as $P = MN$. Without the uncertainties, i.e., $P_n = MN$, $\omega_m = Qn$, which shows that the NRDOB filters out the noise in the high-frequency range.

2) Feedforward Control: Here, feedforward control C^{ff} is designed to transform Z_s , as seen in Table I. Specifically, it is designed to match the second term that is multiplied to $Ks^{-1}MN$ in the denominator. This gives $C_1^{ff} = \frac{N}{C_v}$ and $C_2^{ff} = \frac{N(1+MNC_v)(1-Q)}{C_v}$ for P-PD-PD and NRDOB, respectively. In Table I, the form of a spring port impedance transfer function becomes more intuitive, and this form will be utilized in obtaining the conditions for the maximum rendered stiffness.

In this article, subscripts 1 and 2 denote P-PD-PD and NRDOB, respectively.

III. CLOSED-FORM PERFORMANCE METRICS

This section examines the different passivity conditions at different interaction ports and then develops closed-form expressions of the maximum rendered stiffness and Z -region.

The SEA is passive at a port if its impedance Z attains $|\angle Z| \leq 90^\circ \forall \omega$ or, equivalently, $\text{Re}\{Z\} \geq 0 \forall \omega$. Coupled stability is

then guaranteed if the environment is also (strictly) passive. To perform the coupled stability analysis of the SEA, the SEA and environment can be separated at any of the ports shown in Fig. 1. At the spring port, the environment is E_l , which includes the load dynamics L and the environment E , whereas the environment at the load port is only the environment dynamics E .

The transfer function of the SEA across the interaction ports can be written as an impedance (Y) or an admittance (Z), which are expressed as

$$Y_m = Z_m^{-1} = \frac{\omega_m}{-\tau_s} = MN^2 \quad (6)$$

$$Z_s = \frac{\tau_s}{-\omega_l} = \frac{Ks^{-1}}{1 + Ks^{-1}N^2Y_m} = \frac{Ks^{-1}}{1 + Ks^{-1}MN^2} \quad (7)$$

$$Y_l = Z_l^{-1} = \frac{\omega_l}{-\tau_e} = \frac{L}{1 + LZ_s} = \frac{L(1 + Ks^{-1}MN^2)}{D(s)} \quad (8)$$

at the motor, spring, and load port, respectively.

Often, the passivity of the SEA at the spring port is used as a safety condition [3], [26]. To maintain spring port passivity, the virtual stiffness realized by the impedance control cannot exceed the physical stiffness K [3]. As some systems can safely exceed this limit with real-world safety, passivity can instead be considered at the load port [5], [11], [32], which is less conservative and, thus, used here.

A. Generic Load Port Passivity

The load port impedance can be written as $Z_l(s) = J_l s + B_l + Z_s(s)$; thus, $\text{Re}\{Z_l(\omega)\} = B_l + \text{Re}\{Z_s(\omega)\}$. From Table I, after implementing feedforward control, the spring port impedance can be expressed as

$$Z_s = \frac{K_{imp}}{s} + \frac{(K - K_{imp})(b_p s^p + \dots + b_0)}{a_q s^q + \dots + a_0} \quad (9)$$

where p and q denote the order of the numerator and the denominator, respectively. The real component is

$$\text{Re}\{Z_s(\omega)\} = (K - K_{imp})Z_{re}(\omega) = (K - K_{imp})\frac{N_{re}(\omega)}{D_{re}(\omega)} \quad (10)$$

where $N_{re}(\omega) = (b_0 + (b_2 j\omega)^2 \dots)(a_0 + (a_2 j\omega)^2 \dots) + (b_1 j\omega + (b_3 j\omega)^3 \dots)(a_1 j\omega + (a_3 j\omega)^3 \dots)$, and $D_{re} = (a_0 + (a_2 j\omega)^2 \dots)^2 + (a_1 j\omega + (a_3 j\omega)^3 \dots)^2$. This is found by multiplying the numerator and the denominator by the imaginary conjugate of the denominator and dropping all odd powers of s from the resulting numerator.

Then, applying $\text{Re}\{Z_l(\omega)\} \geq 0$ for $a_0 > 0$ and (10), the load port passivity condition is

$$B_l D_{re}(\omega) + (K - K_{imp})N_{re}(\omega) \geq 0 \quad \forall \omega. \quad (11)$$

The condition of the maximum passive stiffness gain $\bar{K}_{\text{imp}}$ is then the maximum K_{imp} , where (11) holds

$$\bar{K}_{\text{imp}} = \max K_{\text{imp}} \quad \text{s.t. } K_{\text{imp}} \leq K + B_l Z_{\text{re}}^{-1}(\omega) \quad \forall \omega. \quad (12)$$

The maximum stiffness $\bar{K}_{\text{imp}}$ is increased by the load damping B_l , with the factor of $Z_{\text{re}}^{-1}(\omega)$, allowing stiffness higher than the spring port passivity condition.

To eliminate ω from (12), this condition will imposed at the global minimum ω_m of $Z_{\text{re}}^{-1}(\omega)$. An extremum can be determined by equating the first derivative of $Z_{\text{re}}^{-1}(\omega)$ with respect to ω to zero. This extremum is verified to be a minima with the model and parameters used here.

B. Maximum Passive Stiffness

For the P-PD-PD control with ideal differentiation ($C = K_p + K_d s$ and $C_v = P_\omega + D_\omega s$), the coefficients in (9) are $a_0 = KN(P_\omega K_p + N)$, $a_1 = P_\omega N + B_m + KN(P_\omega K_d + D_\omega K_p)$, $a_2 = J_m + ND_\omega(1 + KK_d)$, $b_0 = B_m + NP_\omega$, and $b_1 = J_m + ND_\omega$. The real part of the spring port can be written as

$$Z_{\text{re}}(\omega) = \frac{2\zeta\hat{\omega}\omega^2 + z(\hat{\omega}^2 - \omega^2)}{c(4\zeta^2\hat{\omega}^2\omega^2 + (\hat{\omega}^2 - \omega^2)^2)} \quad (13)$$

where $\hat{\omega} = \sqrt{\frac{a_0}{a_2}}$, $\zeta = \frac{a_1}{2\hat{\omega}a_2}$, $z = \frac{b_1}{b_0}$, and $c = \frac{a_2}{b_1}$. Accordingly, the maximum passive stiffness by (12) is

$$\bar{K}_{\text{imp}} = K + B_l c \frac{4\zeta^2\hat{\omega}^2\omega^2 + (\hat{\omega}^2 - \omega^2)^2}{2\zeta\hat{\omega}\omega^2 + z(\hat{\omega}^2 - \omega^2)^2}. \quad (14)$$

Finding the minima of the right-hand side over all ω by taking the derivative (and verifying the second derivative is positive), the minimum is approximately around $\omega_m = \hat{\omega}$ (rechecked using numerical analysis); then

$$\bar{K}_{\text{imp}} = K + 2B_l\zeta\hat{\omega}c = K + B_l \frac{a_1}{b_1}. \quad (15)$$

The condition for P-PD-PD cases is

$$\bar{K}_{\text{imp1}} = K + B_l \frac{P_\omega N + B_m + KN(P_\omega K_d + D_\omega K_p)}{J_m + ND_\omega}. \quad (16)$$

Recall all these parameters are positive; thus, (16) shows that increasing control parameters can increase the maximum passive stiffness gain.

While increasing the gains P_ω and D_ω increases safe stiffness, these gains are limited by the noise in the motor output. Moreover, a serious practical problem may occur from the double differentiation of a possibly noisy encoder signal.

For a first-order Q -filter in the NRDOB, Z_s is a third-order transfer function with parameters: $\hat{a}_3 = a_2$, $\hat{a}_2 = a_1 + a_2\omega_c + KN^2b_1$, $\hat{a}_1 = a_0 + a_1\omega_c + KN^2b_0$, $\hat{a}_0 = a_0\omega_c$, $b_2 = b_1$, $b_1 = b_0 + b_1\omega_c$, and $b_0 = b_0\omega_c$. After a thorough computation, the maximum passive stiffness condition with the NRDOB is

$$\bar{K}_{\text{imp2}} \leq K + B_l \left(\frac{a_1}{b_1} + \chi \right) \quad (17)$$

where $\chi = KN^2 \frac{b_1(a_1a_2b_0\omega_c + a_0b_1^2) - a_1a_2b_0^2 + a_2KN^2}{a_1^2a_2b_1\omega_c^2 + a_0a_1^2b_1 + a_0a_1b_1^2KN^2 + a_1a_2b_0^2KN^2}$. Using the system and control parameters, $\chi = 1.01e - 6$, which is

still negligible with any value of ω_c . Therefore, when the noise is not considered, the condition in (17) of the NRDOB is equivalent to the case without NRDOB (P-PD-PD case) in (16)

$$\bar{K}_{\text{imp2}} \approx \bar{K}_{\text{imp1}}. \quad (18)$$

According to [19], the increase on the Q -filter order proportionally improves the noise reduction. However, due to the analytical complexity, the effect of the increase in the order of the Q -filter is numerically performed, and the results indicate that the order of the NRDOB does not have significant effect to $\bar{K}_{\text{imp2}}$.

C. Z-Region

Z -width characterizes the range of impedance parameters that can be achieved, originally proposed on haptic systems [13]. For SEAs with inner-loop control, the impedance parameters do not directly determine rendered dynamics—the port dynamics also depend on the inner-loop control. To address this, the Z -region is proposed [8], which measures the range in impedance, which can be rendered at the interaction port—the “depth” of achievable dynamics times the “width” of frequencies over which this change in dynamics can be achieved. With upper and lower achievable impedances Z_u and Z_l , the Z -region is

$$Z^r = \int_{\omega_a}^{\omega_b} W(\omega) |\ln |Z_u(j\omega)| - \ln |Z_l(j\omega)|| d\omega$$

where ω_a and ω_b are arbitrary integration limits and $W(\omega)$ is a weighting filter. The Z -region here is simplified, letting $W(\omega) = 1$, considering the entire frequency domain (i.e., $\omega_a = 0$ and $\omega_b = \infty$), and dropping the outer absolute value (as the upper impedance is typically higher magnitude than the low impedance, except in some cases near the resonance peak)

$$Z^r = \int_0^\infty \ln \left| \frac{Z_u(j\omega)}{Z_l(j\omega)} \right| d\omega. \quad (19)$$

The lower impedance Z_l will be assumed to be obtained at $K_{\text{imp}} = 0$, the zero impedance case that does not pose stability problems for SEAs. The upper impedance Z_u will be assumed to be obtained at the stiffness limits $\bar{K}_{\text{imp}}$, derived in (16) and (18). For the P-PD-PD controller, these simplifications result in the following ratio of impedance $\frac{Z_u}{Z_l} = 1 + \bar{K}_{\text{imp1}}s^{-1}N \frac{M(C_vC+N)}{1+NM C_v}$. Let $L_1(s) = \bar{K}_{\text{imp1}}s^{-1}N \frac{M(C_vC+N)}{1+NM C_v}$ and recall Bode's sensitivity integral

$$\int_0^\infty \ln |(1 + L(s))^{-1}| d\omega = -\frac{\pi}{2} \lim_{s \rightarrow \infty} sL(s) + \pi \sum \text{Re} \{p_k\}$$

where p_k are the right-half-plane poles of L^1 .

As all poles of L_1 are stable (i.e., no right-half-plane poles), the Z -region can be readily evaluated, noting $\lim_{s \rightarrow \infty} sL_1(s) = KNK_dJ_m^{-1}(P_\omega + D_\omega\omega_v)$ and substituting the expression for $\bar{K}_{\text{imp1}}$. The closed-form expression is

$$Z_1^r = KNK_dJ_m^{-1}(P_\omega + D_\omega\omega_v) \left(1 + B_l \frac{\gamma}{J_m + ND_\omega} \right) \quad (20)$$

¹ Assuming $\lim_{s \rightarrow \infty} sL(s) = 0$ is common, but a complete formulation can be found in [33, 3.13].

TABLE II
NOISE TRANSFER FUNCTIONS, WHERE $\tilde{M} = M(1 + N^2 MKs^{-1})^{-1}$, $C = K_p + K_d s$, AND $C_v = P_\omega + D_\omega s \omega_v (s + \omega_v)^{-1}$

	Gain from noise to motor torque, $\frac{\tau_m}{n} =$	High-freq noise gain, α
P-PD-PD	$\frac{-C_v(CNKs^{-1}+1)}{1+C_v(CNKs^{-1}+1)\tilde{M}}$	$(P_\omega + D_\omega \omega_v)(K_d NK + 1)$
w/ NRDOB	$\frac{-C_v CNKs^{-1} + C_v Q - Q P_n^{-1}}{(1+C_v(1-Q)P_n+Q)+\tilde{M}(C_v CNKs^{-1}-C_v Q+Q P_n^{-1})}$	$(P_\omega + D_\omega \omega_v) K_d NK$

where $\gamma = K^{-1}P_\omega N + K^{-1}B_m + NP_\omega K_d + ND_\omega K_p$. If model uncertainty and noise are ignored, the NRDOB case attains the same port dynamics, so $L_2 = L_1$; then

$$Z_2^r = Z_1^r. \quad (21)$$

In the next section, the conditions for the maximum stiffness and the Z -region are redefined considering the noise analysis.

IV. NOISE ANALYSIS AND CONTROLLER SYNTHESIS

This section presents the noise sensitivity and examines performance normalized by noise sensitivity. Then, the controller synthesis is presented based on noise sensitivity.

A. Noise Analysis

A major practical limitation of SEA control is due to high-frequency noise (e.g., in strain-gauge-based SEAs) or quantization/discretization effects (e.g., in encoder-based SEAs), which limit controller gains. Excessive noise in the motor input signal can saturate the input, increase power consumption, accelerate drivetrain wear, and cause audible/tactile noise, which is undesired in some applications.

This section analyzes the closed-loop gain from a noise signal n to motor input τ_m . This noise, which enters the motor output measurement as seen in Fig. 2, models quantization or electrical noise. The closed-loop transfer functions from n to τ_m can be written for each architecture, shown in Table II. Note that the block diagram does not exactly correspond, as noise does not enter the physical feedback of τ_s .

To compare the noise performance of the different controllers, their high-frequency noise gain can be compared. The closed-form expression for this gain can be found as frequency $\omega \rightarrow \infty$,

$$\lim_{\omega \rightarrow \infty} \left| \frac{\tau_m(j\omega)}{n(j\omega)} \right| = \alpha \quad (22)$$

where closed-form expressions for α of the controllers are shown in Table II. An increase in α corresponds to a uniform increase in magnitude of $n \rightarrow \tau_m$ above 80 Hz for controller parameters used here, so it successfully describes a range of noise.

This high-frequency noise gain α will be used to compare between controllers—many controllers offer better performance, but may have unacceptable noise characteristics. Normalizing by noise gain can allow a more fair comparison. In the following, it will be assumed that there exists a high-frequency noise gain limit, largely set by the hardware (depending on encoder resolution, sampling rate, etc.), which must be respected by any controller. This hardware limit can be denoted by $\bar{\alpha}$, where any controller must keep $\alpha < \bar{\alpha}$.

B. Noise Limited Performance

Considering the noise sensitivity, new closed-form expressions are presented. This analysis considers a LPF on the derivative action of the velocity loop, which becomes $C_v = P_\omega + D_\omega \omega_v s(s + \omega_v)^{-1}$.

1) **Maximum Stiffness:** To normalize performance by the high-frequency noise gain, the maximum passive stiffness from (16) to (18) can be rewritten with α , as

$$\bar{K}_{\text{imp1}} = K + B_l \frac{B_m + \alpha_1 + P_\omega(N-1) - D_\omega \omega_v}{J_m + ND_\omega} \quad (23)$$

$$\bar{K}_{\text{imp2}} = K + B_l \frac{B_m + \alpha_2 + P_\omega N}{J_m + ND_\omega}. \quad (24)$$

The improvement in $\max K_{\text{imp}}$ of the NRDOB over P-PD-PD when $\alpha_1 = \alpha_2$ is $\frac{B_l}{J_m + ND_\omega} (P_\omega + D_\omega \omega_v)$, which using the parameters in Section V, and (27) evaluates to an improvement of 2.4×10^5 (N/m), i.e., $1.7K$. Intuitively, if the noise reduction allows the gains P_ω and D_ω to be further increased, it can be clearly seen in the above equations that increasing these gains will allow higher maximum passive stiffness.

2) **Z -Region:** To compare the Z -region of the different architectures, substituting the high-frequency gain α , as seen in Table II, into the expression for the Z -region from (20) and (21), dropping the $\pi/2$ term for comparison

$$Z_1^r = (\alpha_1 - P_\omega - D_\omega \omega_v) J_m^{-1} \left(1 + B_l \frac{\alpha_1 + B_m + NP_\omega}{\tilde{J}_m K} \right) \quad (25)$$

$$Z_2^r = \alpha_2 J_m^{-1} \left(1 + B_l \frac{\alpha_2 + B_m + NP_\omega}{\tilde{J}_m K} \right) \quad (26)$$

where $\tilde{J}_m = J_m + ND_\omega$. From these expressions, it can be shown that the NRDOB outperforms the P-PD-PD for positive control gains.

C. Controller Synthesis

With the performance metrics written in closed form (16) and (20), they can be used for controller synthesis. A major challenge in design of cascaded control systems is a relatively high number of parameters, which limits the efficacy of intuitive or iterative gain tuning. While an optimization-based approach can handle the complexity, when the optimization is not constrained, the derivative action (of either outer or inner loop) can be increased arbitrarily large to realize arbitrarily good performance on SEAs. Here, the controller gains are constrained such that $\alpha < \bar{\alpha}$, where $\bar{\alpha}$ is found experimentally on this hardware to be $\bar{\alpha} = 0.18$.

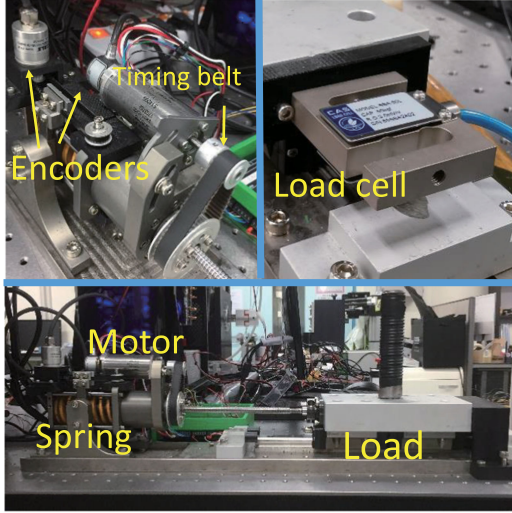


Fig. 3. RFSEA for high-stiffness control performance verification.

The controller gain optimization problem for the NRDOB is written as

$$\arg \max_{K_p, K_d, P_\omega, D_\omega} \lambda \bar{K}_{\text{imp}2} + Z_2^r$$

such that $(P_\omega + D_\omega \omega_v) K_d N K \leq \bar{\alpha}$.

$\bar{K}_{\text{imp}2}$ in (24) and Z_2^r in (26) are nonconvex functions of the control parameters. The parameter λ is a weighting term for the relative importance of the Z -region and maximum stiffness. The inequality constraint limits the high-frequency noise gain and can be eliminated by assuming that the inequality attains equality (where Z -region performance is best) and using substitution to eliminate D_ω as $D_\omega \omega_v = (K_d N K)^{-1} \bar{\alpha} - P_\omega$.

Using the `fmincon` command in MATLAB with an initial position of $[K_p, K_d, P_\omega, D_\omega] = [0.5, 0.05, 2.0, 0.005]$ and bounds $[K_p, K_d, P_\omega, D_\omega] \leq [0.5, 0.05, 4.0, 0.01]$, the optimization with $\lambda = 0.8$ and $\omega_v = 157$ returns a result of

$$[K_p, K_d, P_\omega, D_\omega] = [0.5, 0.0024, 4.0, 0.01]. \quad (27)$$

The optimized gains decrease K_d and increase in inner-loop gains, resulting in an theoretical increase in $\bar{K}_{\text{imp}2}$ of 1.09 relative to the initial gains. This improvement is validated experimentally in the next section. The gain limits are required to address hardware constraints (e.g., the simplified noise model does not capture all sources of noise), and are found empirically.

V. EXPERIMENTAL VERIFICATION

A reaction-force series elastic actuator (RFSEA) is used, as shown in Fig. 3. It consists of brushless dc motor (Maxon EC four-pole 305015) with a maximum continuous torque of $92.9 \text{ mN} \cdot \text{m}$. The motor is equipped with an incremental encoder (SCANCON, 20000 counts/rev) before the gearbox, while the spring deflection is obtained by the high-resolution rotary incremental encoder (RLS RE22IC, 13 bit). The motor velocity is obtained through numerical differentiation of motor position, with the derivative LPF cutoff at $\omega_v = 157$.

The physical parameters of the RFSEA are identified by using the fast Fourier transform analyzer (ONO-SOKKI, CF-9400),

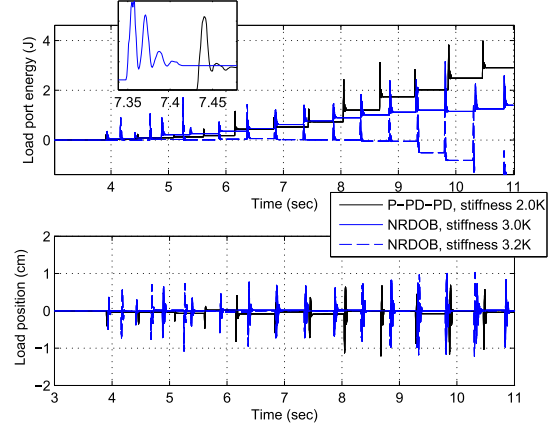


Fig. 4. Total energy flow at load port under impact excitation, showing that the NRDOB is passive at $3K$, and not at $3.2K$.

which are $J_m = 6.4 \times 10^{-6} \text{ kg} \cdot \text{s} \cdot \text{m}^2$, $B_m = 0.6 \times 10^{-4} \text{ N} \cdot \text{m} \cdot \text{s/rad}$, $M_l = 41.3 \text{ kg}$, $B_l = 100 \text{ N} \cdot \text{s/m}$, $K = 141350 \text{ N/m}$, and $N = 1.27 \times 10^{-4}$.

The Q -filter of the NRDOB is a first-order LPF, and a cut-off frequency of 40 Hz was chosen as the highest frequency, which did not compromise stability in a range of controllers and environments.

A. Maximum Passive Stiffness Gain Evaluation

The passivity of a system is not easy to evaluate in a real experiment. Here, the passivity observer (PO) is utilized to determine the passivity violation by measuring the energy at a specific port (here, load port) [5], [34]. Using the PO, negative energy means passivity violation and thus not safe according to the load port condition. For the load port passivity, the port energy flow from time 0 to t_p is $E_l = \int_0^{t_p} \tau_e(t) \omega_l(t) dt$. This validation has been previously established as a practical condition for stability in contact with a high-stiffness environment [11].

In these experiments, a hammer is used to strike the static load side, where a load cell was installed to measure τ_e . The result of this experiment is shown in Fig. 4. The P-PD-PD controller is passive at $2.0K$ but marginally passive at $2.1K$. The NRDOB with optimized gains in (27) is passive at $3.0K$, where the NRDOB stiffness $3.2K$ is marginally passive. A detail inset shows that the impact is captured with sufficient resolution that estimating the energy flow during impact is reasonable. The hand-tuned gains for the NRDOB of $[K_p, K_d, P_\omega, D_\omega] = [0.5, 0.005, 2.0, 0.01]$ provide a passive limit of $2.5K$ in the same test methodology, showing that the optimized gains improve the passive limit.

The impact of the NRDOB on the limit of D_ω was tested by increasing D_ω to the empirical noise limit (all other parameters same). With the NRDOB, $D_\omega = 0.14$ is possible, while the P-PD-PD had automatic heat shutoff at $D_\omega = 0.03$.

A comparison with an LPF on the P-PD-PD controller is shown in Fig. 5(top), where cutoff frequency ω_c of the LPF and the Q -filter varied in hammer strike tests; all other control parameters are same between NRDOB and P-PD-PD. The NRDOB is passive (positive average port energy) in any value of ω_c while with the LPF becomes nonpassive when ω_c is 10 and 20 Hz:

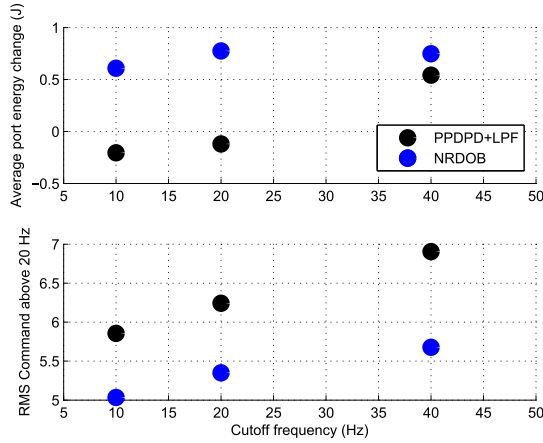


Fig. 5. Effect of cutoff frequency of the LPF (P-PD-PD) and the Q -filter (NRDOB) on the average energy flow (negative is not passive) and RMS high-frequency motor torque (above 20 Hz).

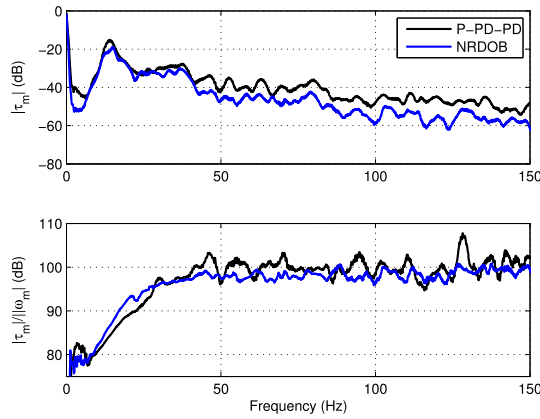


Fig. 6. Motor command in the frequency domain at $K_{\text{imp}} = 2K$ in hammer strike experiments, as well as gain $\omega_m \rightarrow \tau_m$.

the passivity of the NRDOB is less affected by ω_c . As described in Section IV-A ($n \rightarrow \tau_m$), the high-frequency (>20 Hz) RMS motor command is also compared in Fig. 5(bottom)—reducing ω_c reduces noise for both systems, and the NRDOB has less high-frequency motor noise than the LPF.

A comparison of the motor command in the frequency domain, at the same controller gains in the hammer strike experiments, can be seen in Fig. 6 (LPF and Q -filter cutoff frequency 40 Hz). The NRDOB has a lower magnitude of the motor input, especially above 40 Hz, as well as a lower gain from ω_m to τ_m .

B. Closed-Loop Performance Evaluation

The position control performance is evaluated with step and sine load position references, with a comparison in Fig. 7 with gains $K_p = 0.5$, $K_d = 0.005$, $P_\omega = 2$, and $D_\omega = 0.01$, and K_{imp} is at the safe limit, to show the best position performance achievable on the two systems. The step response (see Fig. 7, top) is done with no payload and a 5-kg payload, where the NRDOB shows a faster settling time and less steady-state error, reflecting the NRDOB's higher stiffness and compensation of friction/model uncertainties. In the sine tracking case (which includes environment contact at a positive load position), the

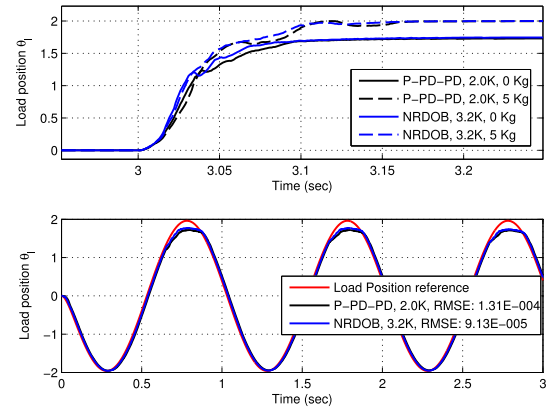


Fig. 7. Position tracking comparison of the NRDOB and P-PD-PD, step and sine reference for load position. The NRDOB can achieve a higher stiffness and, therefore, lower RMSE tracking error and faster settling time.

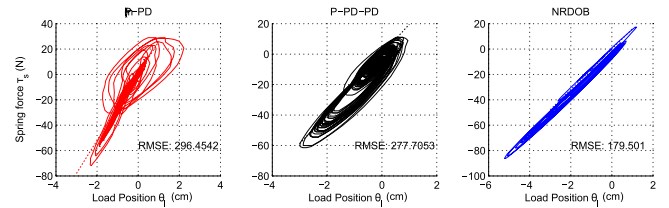


Fig. 8. Rendered stiffness at spring port, stiffness of $2K$, data from sine tracking with payload without environment contact.

NRDOB again achieves a slightly lower tracking root-mean-square error (RMSE) of 9.13×10^{-5} compared with the P-PD-PD: 1.31×10^{-4} .

In Fig. 8, a final comparison shows the accuracy of the rendered stiffness (load position versus spring torque) achieved during the sine tracking comparisons at a stiffness $K_{\text{imp}} = 2.0K$. The NRDOB achieves a much lower RMSE from the desired stiffness.

VI. CONCLUSION

This article derived closed-form expressions for the highest passive stiffness an SEA can render. This upper limit is shown to strongly depend on the high-frequency noise gain of the controllers, which sets a practical limit on performance—i.e., derivative action helps, but cannot be set arbitrarily high. This high-frequency noise gain can be reduced with the NRDOB, which is shown to improve performance in safe stiffness and Z -region, and closed-form expressions for maximum stiffness and Z -region used to find optimized gains. The system was validated, achieving load port passivity at up to 3.0 times the physical stiffness on the RFSEA here, a value that would vary between hardware systems.

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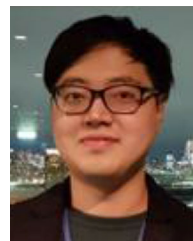


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